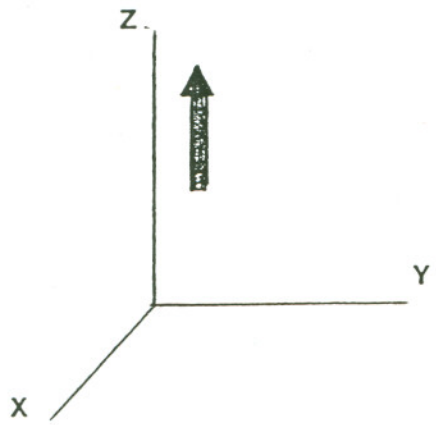
PHYSICAL CHEMISTRY CUMULATIVE EXAM

April 26, 1997

Nuclear magnetic resonance (NMR) spectroscopy is arguably the most important and widely used analytical technique the has chemist has to study structure, function and dynamics of molecules and molecular systems. The power of the technique can be appreciated in that one is able to obtain in a rather routine manner spectra of solids, liquids and gases. There are also approximately 400 pulse sequences available to the chemist to obtain 1D, 2D, 3D and 4D spectra that can be used to identify a molecule, determine its 3 dimensional structure, or study its motion, for example. Perhaps, because of the many pulse sequences available that perform so many functions, NMR spectroscopy has become a "push-button" spectroscopy. While this is very useful, it means that most students never really know what happens to the molecular system of interest in the probe once the buttons are pushed and have little appreciation or understanding of how they get the spectra that they want.

Suppose that the system of interest to you has only one nucleus that has a spin quantum number greater than zero and that there is no spin-spin coupling in the system. Consider the following two pulse sequences. Using the vector diagram method show where the net magnet moment is in the xyz frame-work notation shown below (the direction of the applied magnetic field is along the z axis and the transmitter/receiver coil is positioned along the y axis) after each pulse and delay and draw the signal that would result from the Fourier transform of the acquired free induction decay. In the xyz frame-work shown the net magnetic moment of the nucleus at the beginning of the experiment is along the direction of the z axis and is denoted by the arrow.



PULSE SEQUENCE 1:

delay - $(180^\circ \text{ pulse})_x$ - τ - $(90^\circ \text{ pulse})_x$ - acquire free induction decay

In this pulse sequence, τ, is a delay that can be varied from one experiment to another and x refers to the axis along which the RF is applied. Draw the vector diagram of the spin system after each pulse and the final NMR spectrum that one obtains using a very short and then a very long value of τ (i.e., these are two separate experiments).

PULSE SEQUENCE 2:

delay - $(90^\circ \text{ pulse})_x$ - τ - $(180^\circ \text{ pulse})_y$ - τ - acquire free induction decay

In this pulse sequence, both τ's, are delays that have the same value in a given experiment. The 90° pulse is applied along the x axis and the 180° pulse is applied along the y axis. Draw the vector diagram of the spin system after each pulse and after the end of the second τ delay. Also draw the NMR spectrum that one obtains from the Fourier transform of the free induction decay.

Now consider a heteronuclear spin system that is to be studied by Pulse Sequence 3.

PULSE SEQUENCE 3:

Consider a two spin system composed of a single ^{13}C - ^{1}H moiety. One will experimentally observe the ^{13}C nucleus. Draw the vector diagram of the ^{13}C spin system after each pulse and after the second τ delay. All of the τ values are identical in both the ^{13}C and ^{1}H pulse sequences. The 180° pulses for ^{13}C and ^{1}H occur at the same time.

^{13}C

delay - $(90^\circ \text{ pulse})_x$ - τ - $(180^\circ \text{ pulse})_y$ - τ -acquire FID

^{1}H

$(180^\circ \text{ pulse})_x$ - τ - broadband decouple